Faculty of Computing

IE1030 – Data Communication Networks

Year 1 Semester 1 (2024)

Network Design Assignment

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1.INTRODUCTION

Tech World INC..A small-scale company that provides software development and IT services. The company is now establishing a new office building which requires modern, Scalable and secure infrastructure. The office building contains three floors with a total area of 30000 square feet, and with 80 employees across various departments such as Administration, Sales, IT, HR and Development.

This report details the network design for a business organization based on the allocated IP range of 192.168.93.0/24. The goal pf the assignment is to implement an efficient IP addressing by dividing the network into subnets using VLANs.

1.1Company overview and network requirements

Company Overview:
• Company Name: TechWorld Inc.
• Industry: Software Development and IT Services
• Number of Employees: 80
• Office Location: 3-floor building with 10,000 square feet of office space per floor.

Business Requirements:

1. Departments and Workstations:
   o Administration Department: 15 computers, 2 printers, 1 scanner.
   o Sales Department: 25 computers, 3 printers, 1 copier.
   o IT Department: 10 computers, 1 server, 1 printer.
   o Development Team: 20 computers, 2 servers, 1 shared high-performance storage device.
   o HR Department: 5 computers, 1 printer.
   o Meeting Rooms (3 rooms): Each room requires a wireless access point and video conferencing equipment.

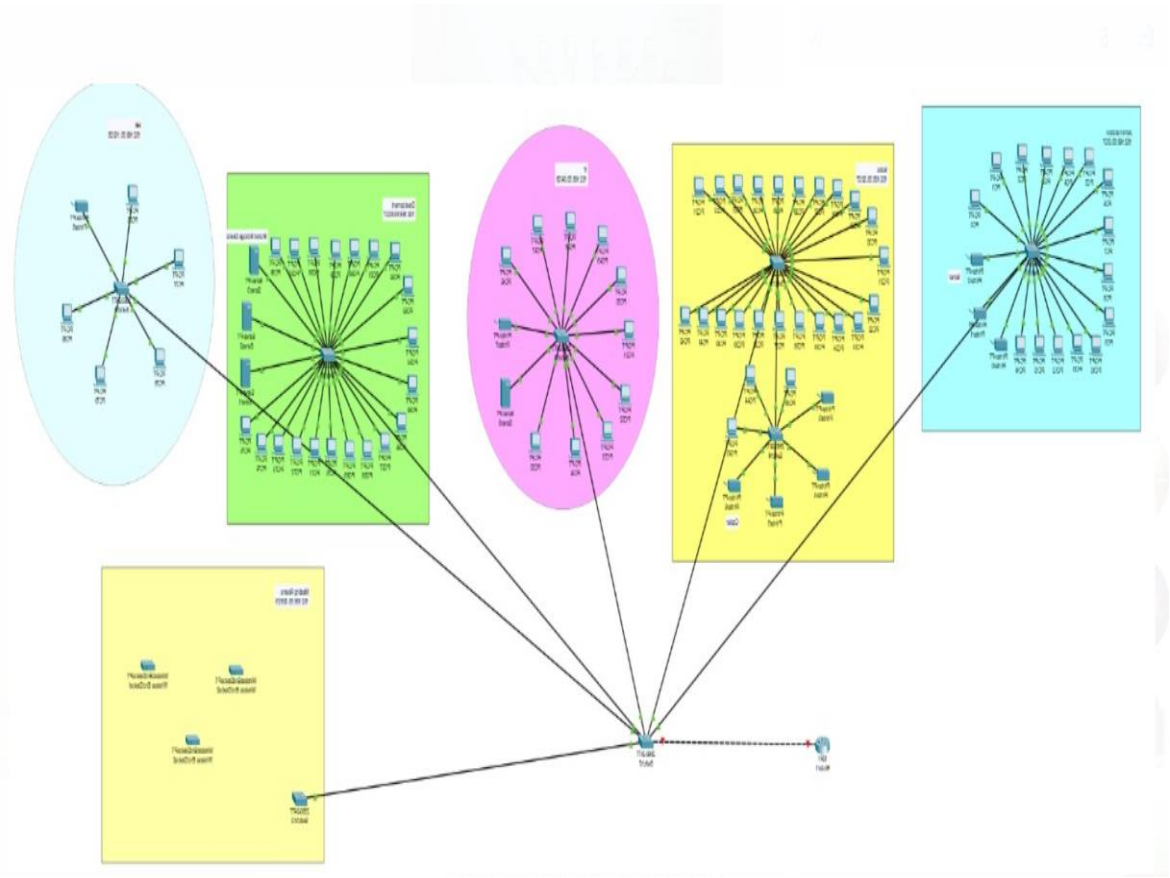
2. Network Services:
   o Internet Connectivity: High-speed broadband connection.
   o Internal Communication: Email server, file sharing, and internal chat service.
   o External Communication: Web server hosting the company website and a secure VPN for remote workers.

3. Security Requirements:
   o Data Protection: Secure sensitive company data with encryption.
   o Network Segmentation: Implement VLANs to separate network traffic between different departments.
   o Access Control: Use firewalls and access control lists (ACLs) to protect the network.
   o Wireless Security: Implement secure wireless networks for employees and guests.

4. Scalability and Flexibility:
   o the network should support the company's growth plans, including the addition of more employees and new departments.
   o Provide solutions for adding more devices without major reconfigurations.
   o Plan for future integration of IoT devices such as smart office equipment.

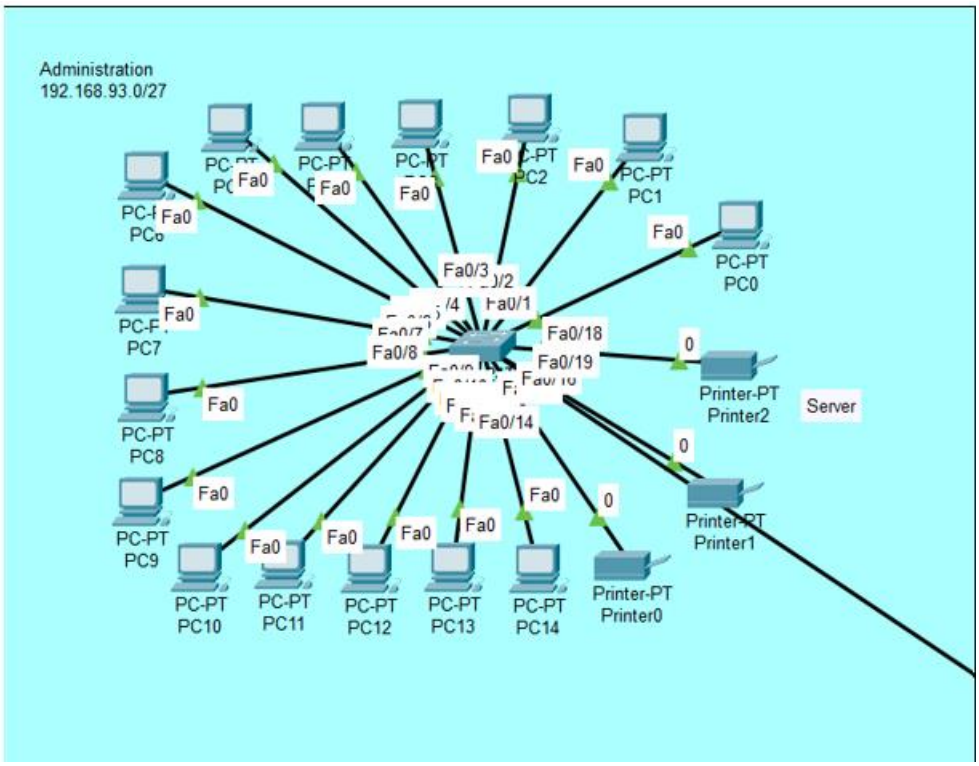
5. Budget:
   o the company has allocated a budget for the entire network setup, including hardware, software, and installation.

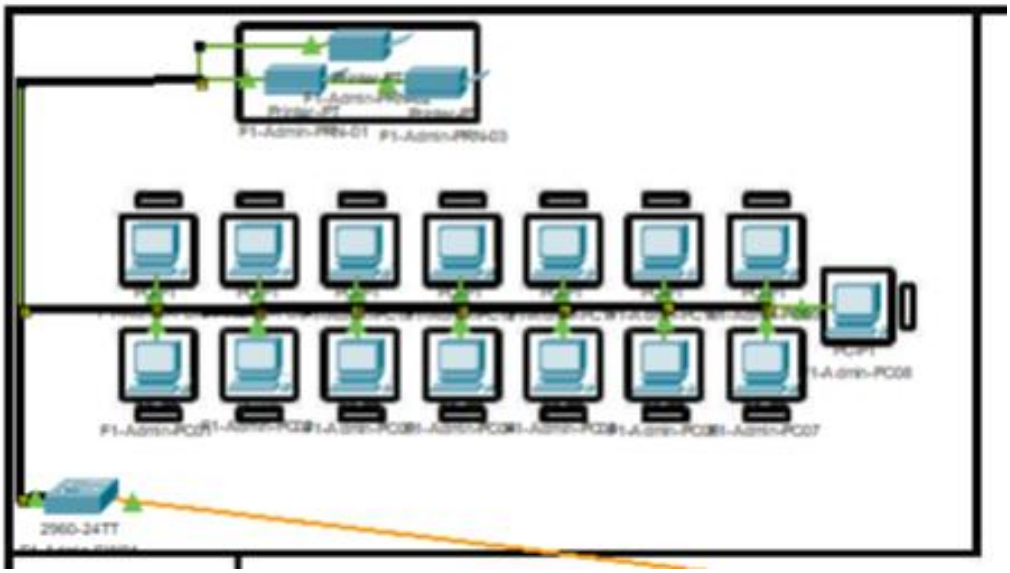
2.NETWORK TOPOLOGY DESIGN



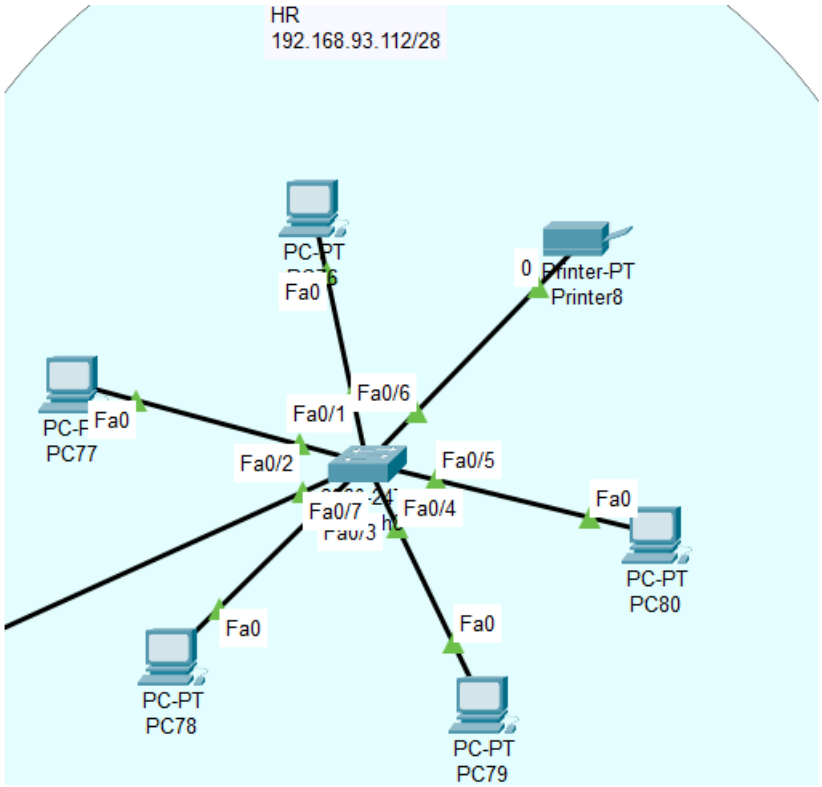
2.1 First floor logical view and Physical view

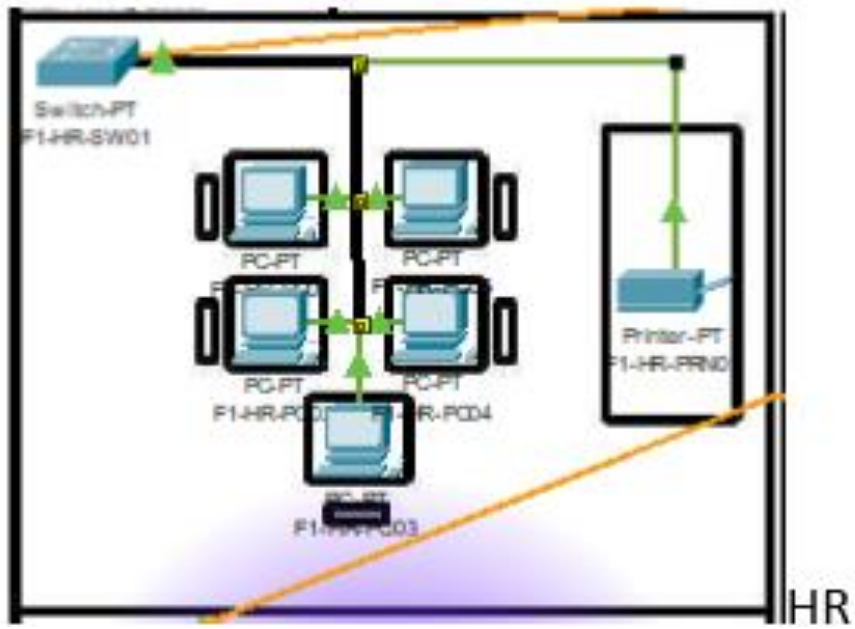
- Administration



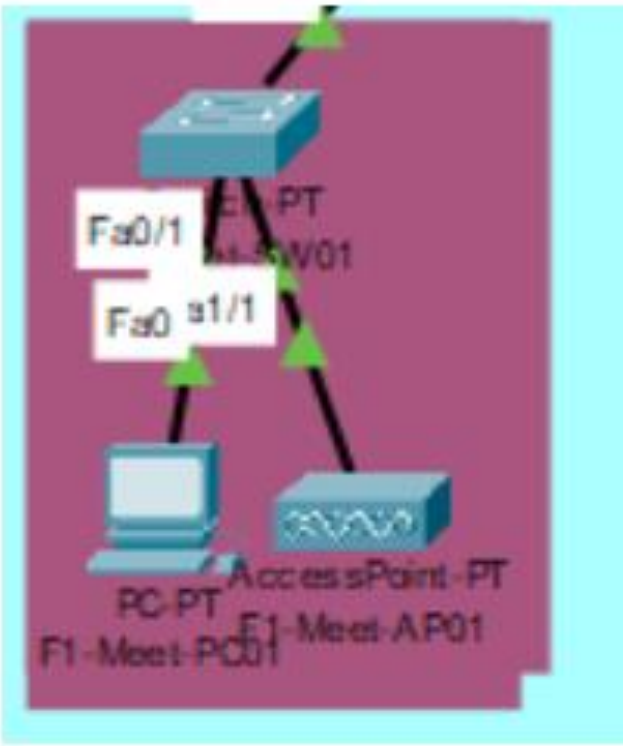


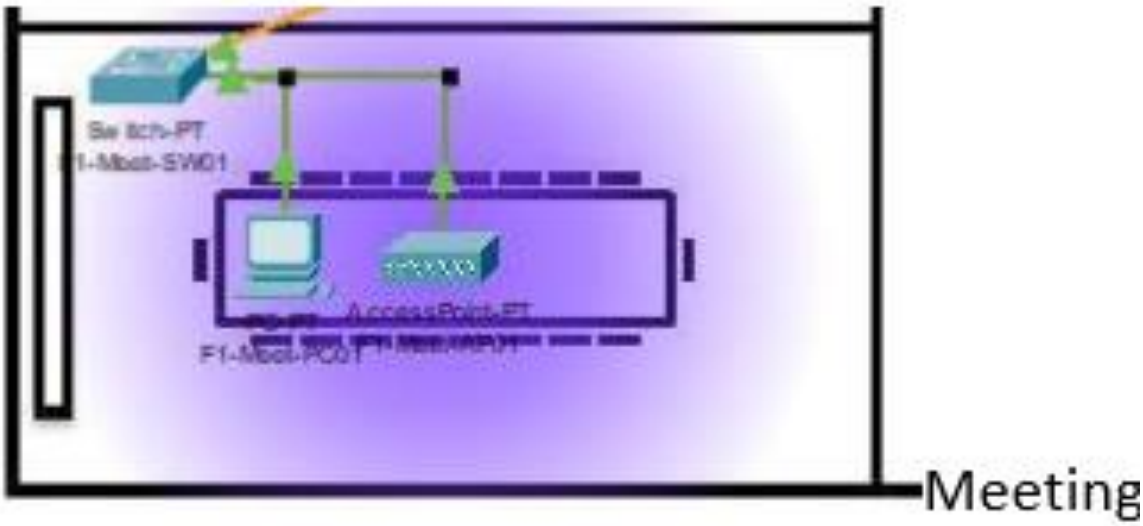
➢ HR Department





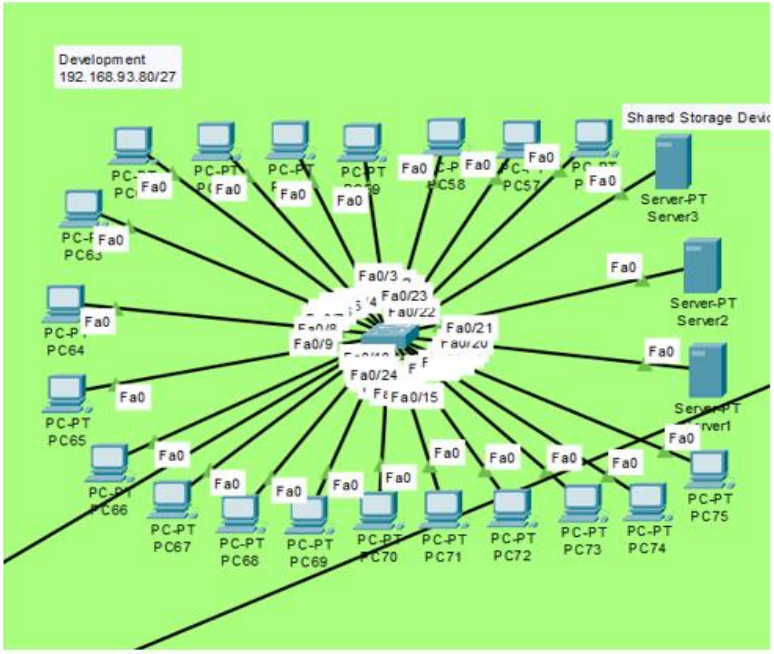
- Meeting Room

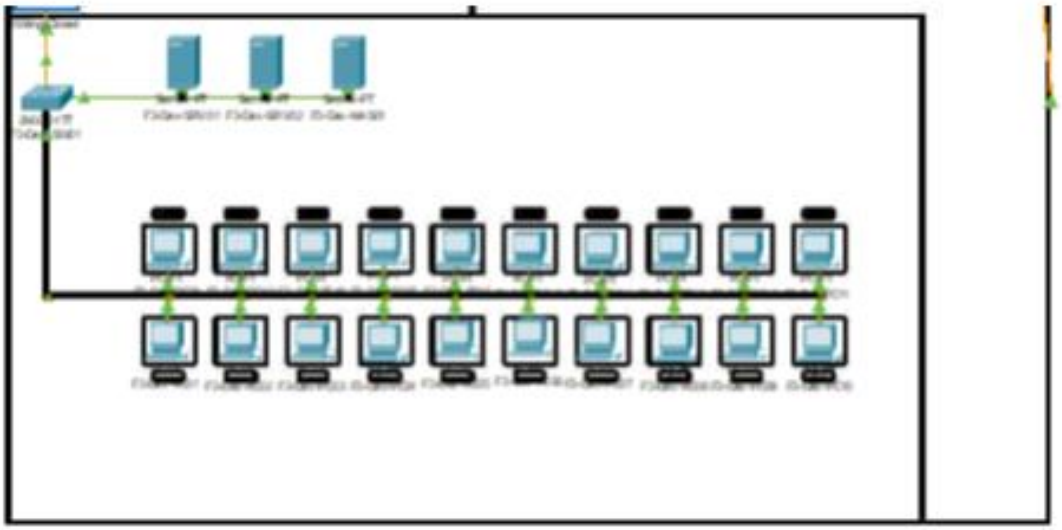




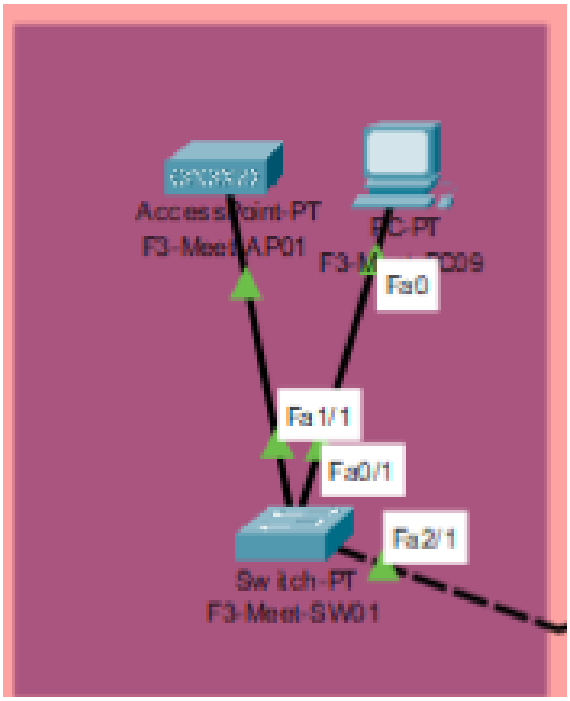
2.2 Second floor logical view and physical view

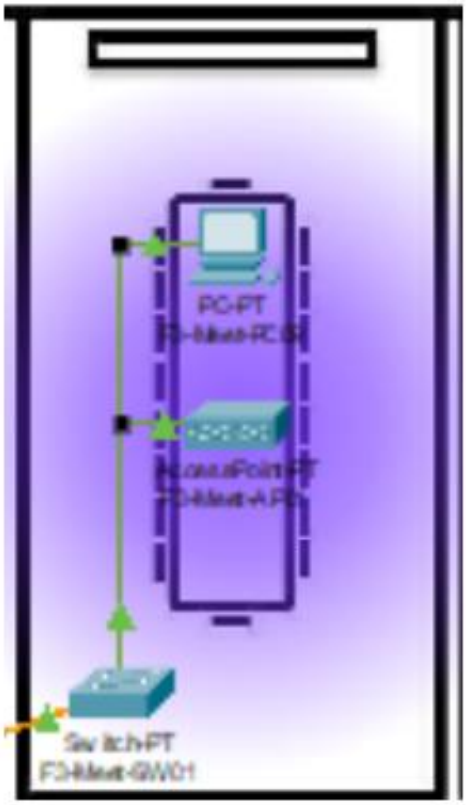
- Development Team





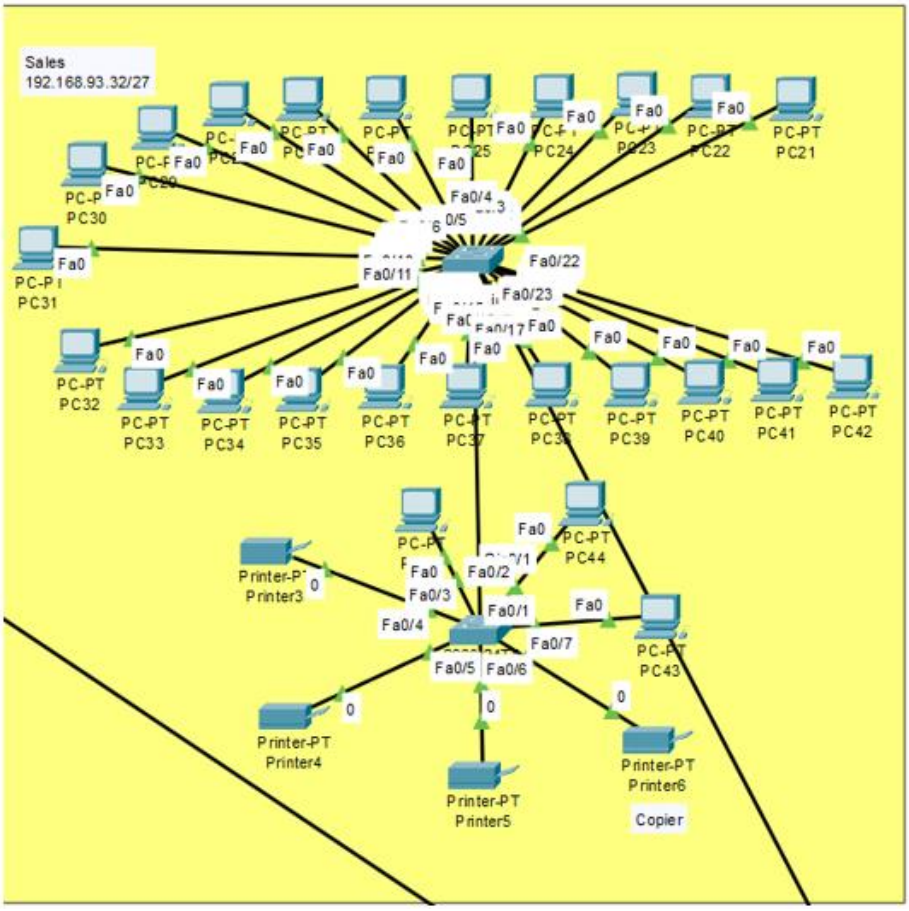
- Meeting Room

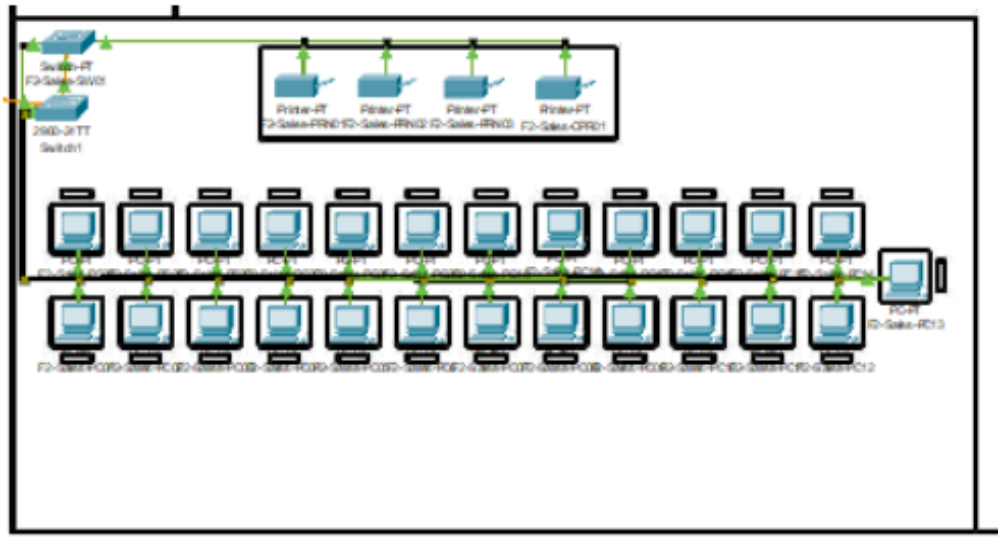




2.3 TIRD FLOOR PHYSICAL AND LOGICAL VIEW

- Sales Department





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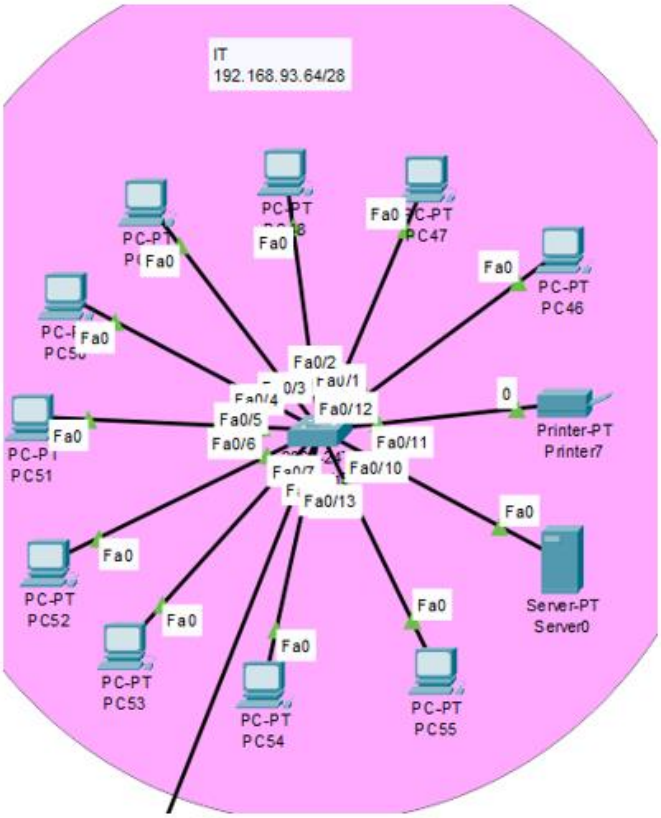
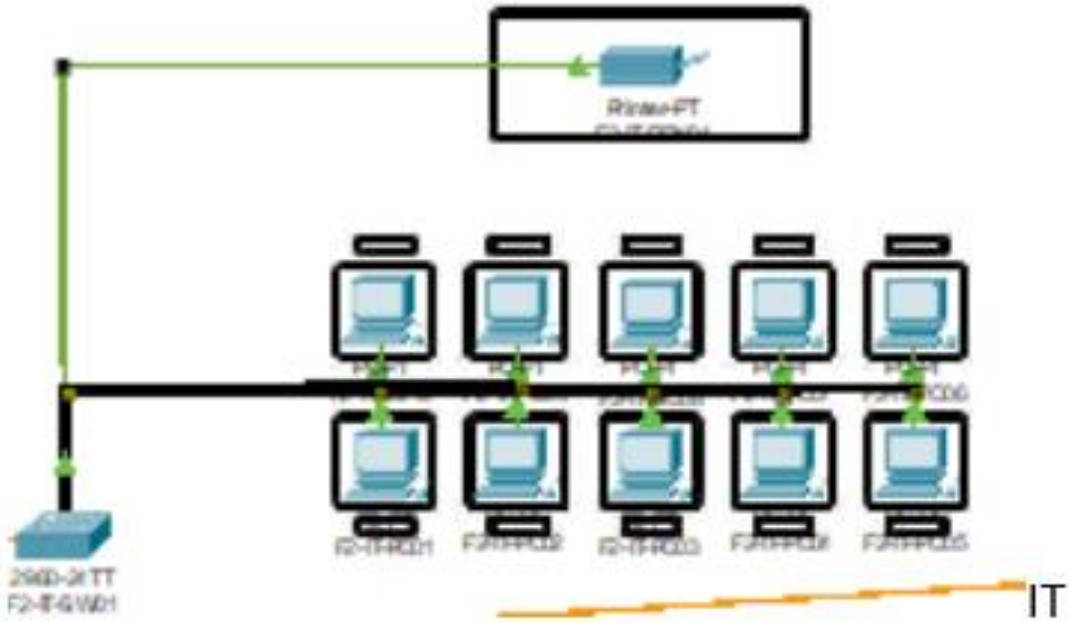
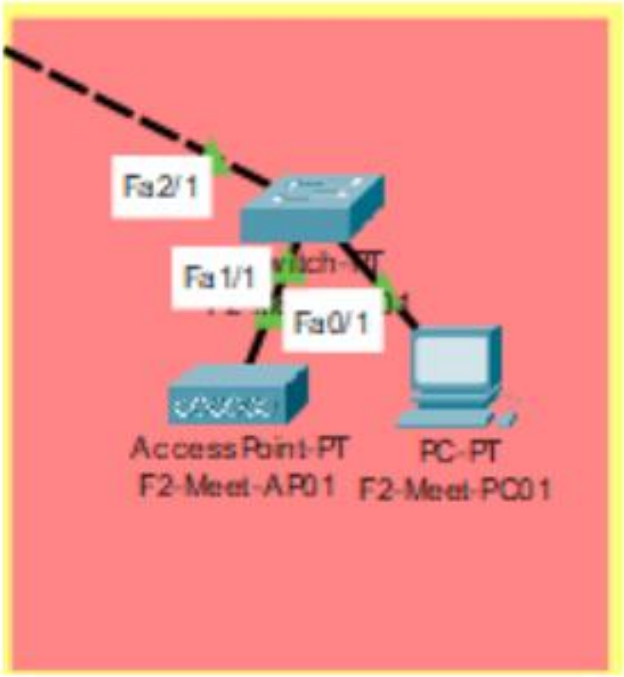
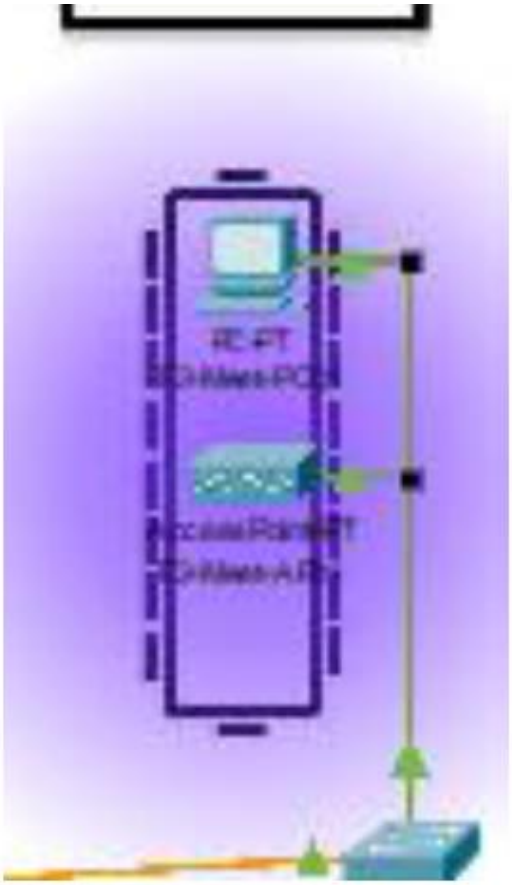
- IT Department





- Meeting Room





IP Addressing Scheme

Department/Area	Devices	Subnet	Assigned IP Range	Specific IP Allocations
Administration	15 Computers 2 printers 1 Scanner	192.168.93.0/27	192.168.93.1 To 192.168.93.30	Router: 192.168.93.1 Computers: 192.168.93.2-16 Printers: 192.168.93.17-18 Scanner: 192.168.93.19
Sales	25 Computers 3 Printers 1 Copier	192.168.93.32/27	192.168.93.33 to 192.168.93.62	Router: 192.168.93.33 Computers: 192.168.93.34-58 Printers: 192.168.93.59-61 Copier: 192.168.93.32
IT Department	10 Computers 1 Server 1 Printer	192.168.93.64/28	192.168.93.65 To 192.168.93.78	Router: 192.168.93.65 Computers: 192.168.93.66-75 Server: 192.168.93.76 Printer: 192.168.93.77
Development Team	20 Computers 2 Servers 1 Storage	192.168.93.80/27	192.168.93.81 to 192.168.93.110	Router: 192.168.93.81 Computers: 192.168.93.102-103 Storage: 192.168.93.104
HR Department	5 Computers 1 Printer	192.168.93.112/28	192.168.93.113 To 192.168.93.16	Router: 192.168.93.113 Computers:

				192.168.93.114-118 Printer: 192.168.93.119
Meeting Rooms	3 WAPs 3 video conferencing systems	192.168.93.128/29	192.168.93129 To 192.168.93.136	WAPs: 192.168.93.129-131 Video systems: 192.168.93.132-134
Network Infrastructure	Routers Switches Firewall	192.168.93.192/29	192.168.93.193 To 192.168.93.200	Router: 192.168.93.193 Main Switch: 192.168.93.194 Firewall: 192.168.93.195

VLAN configurations

1.Administration Department:
VLAN – 10
2.Sales Department
VLAN – 20
3.Development
VLAN – 30
4.IT department
VLAN – 40
5.HR Department
VLAN - 50

BUDGETING

ITEM	Model/Brand	Quantity	Unit price (LKR)	Total Price (LKR)
Router	CISCO ISR 4431	1	350,000	350,000
Switch (24 port)	CISCO Nexus 9300	3	250,000	750,000
Access Points (wi-fi 6)	Ubiquiti UniFi 6 Long Range	6	55,000	330,000
Firewall	Fortnite Fortnite Gate 100 E	1	275,000	275,00
Servers	Dell PowerEdge R540	3	450,000	1,350,000
High Performance Storage Device	Dell EMC Islion	1	750,000	750,00
Desktop Computers (Development)	Dell precision 5820	20	275,000	5,500,000
Desktop Computers (Admin and Sales)	Dell OptiPlex 3080	55	90,000	4.950,000
Printer	HP LaserJet Enterprise MFP	5	85,000	425,000
Video Conferencing Equipment	Polycom RealPresence Trio	3	320,000	960,000
Wireless Controller	CISCO Wireless controller 5520	1	350,000	350,000
Cabling and accessories	-	-	500,000	500,000
Uninterruptable Power supply	APC-Smart UPS	4	175,000	700,00
Backup Servers	Dell PowerEdge R540	2	450,000	900,000
RAID System	Dell Power Vault ME4024	1	900,000	900,000
Installation Ans Setup Fees	-	-	1,500,000	1,500,000

SECURITY CONSIDERATION

For the security considerations in the Tech world INC, we can use logical and physical security measures. Security is a very crucial component. Here are some recommended security measures along with the business requirements.

1.Network segmentation with the use of VLANS

- Can create VLANS to reduce the network traffic between different departments. Also, this improves the security by preventing the unauthorized access.

Example: VLAN 10 for administration
VLAN 30 for IT

2.Installing Firewalls in the network

- By installing a firewall in between the networks can also prevent the unauthorized access to the network.

3.Providing physical security

- Providing physical security is also plays a very important role in the security measures.
- Restricting the access to the servers, routers, switches by keeping them in locked rooms.
- The use cctv cameras also helps to prevent any physical unauthorized access to the important devices

4.Regular software updates and backups.

- It is very important to ensure that all the network devices have updated firmware and security patches.

5.Wirelsess Security

- Use WAP3 encryption to secure the wireless communication
- By implementing SSID separation can create different SSIDs for employees and guests and it ensures that they don’t have access to the same resources.

6.Data Encryption

- All the sensitive data records in the company should be kept encrypted.
- Enabling full-disk encryption for all servers, especially for the high-performance storage device used by the development team#

7.Mail Server Security

- Enable spam filtering can detect to block malicious and unrequested mails
- Also the use of a strong password policy is very necessary for the company to prevent unauthorized access to the email accounts.

8.Endpoint Security

- Antivirus and Anti-malware solutions: Ensure all endpoints (PCs, servers) have up-to-date antivirus and anti-malware software installed

SCALABILITY AND FLEXIBILITY

As the company grows with time the current IP address pool will not meet up the company's requirements. So in order to solve this problem we can use options such as,

1.The use of a larger IP network range
2.The use Dynamic IP addressing with DHCP

- At the moment the network limited to 192.168.93.0/24 which provide 256 IP addresses. But when the company grows this won't be suuficient.
- The use of DHCP simplifies the process of adding new devices to the network.